CSE Teaching Materials — Table of Contents

Civilisational Systems Engineering | Infrastructure Academy

Updated: June 2026 **Status:** Complete with TP-064 integration

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FOLDER STRUCTURE

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MATERIAL ORGANIZATION BY TIER

TIER 1: FOUNDATIONAL (Start Here)

Recommended for: First-time learners, educators new to CSE

1. **La Menara - The 12 Remarkable Relays** (Part 1 Teaching Deck)
   - Duration: 45–60 minutes
   - Focus: Introduction to the 12 Relays and their civilisational significance

- Key concepts: Fire, Tree, River, Stone, Road, Metal, Wheel, Engine, Orbit, Web, Code, Torus
- Outcome: Understanding of relay structure and progression

2. **CSE Framework Overview - Video 1** (10 seconds)
   - Quick visual introduction to CSE principles
   - Best used: Before or after Part 1 deck
   - Outcome: Rapid conceptual overview

3. **CSE Foundational Paper**
   - Duration: 30–45 minutes
   - Focus: Theoretical foundations of Civilisational Systems Engineering
   - Key concepts: System definition, relay mapping, knowledge transmission
   - Outcome: Theoretical grounding in CSE methodology

TIER 2: APPLIED (Build Understanding)

Recommended for: Students, professionals, educators with Tier 1 foundation

1. **Part 2: Civilisational Systems Engineering — 37 Base Case Examples**
   - Duration: 90–120 minutes
   - Focus: Real-world infrastructure applications across all 12 relays
   - Key concepts: Case studies, implementation patterns, relay-specific challenges
   - Outcome: Practical understanding of CSE in context

2. **CSE Application - Video 2** (10 seconds)
   - Demonstration of CSE framework in practice
   - Best used: After Part 2 deck
   - Outcome: Visual reinforcement of applied concepts

3. **UVM-DF Narrative Manuscript**
   - Duration: 45–60 minutes
   - Focus: Unified Value Model and Dearden Field narrative
   - Key concepts: Value integration, field theory, narrative methodology
   - Outcome: Understanding of value systems in CSE

TIER 3: ADVANCED (Master the System)

Recommended for: Researchers, advanced practitioners, system designers

1. **Hyper Learning Edutainment**
    - Duration: 60–90 minutes
    - Focus: Educational methodology combining CSE with immersive learning
    - Key concepts: Spacing effect, remembrance architecture, pedagogical engineering
    - Outcome: Mastery of learning system design

2. **TP-064: What Ebbinghaus Could Not Build, iAAi Can** (COMPLETE)
    - Duration: 120–150 minutes
    - Focus: Theoretical integration of memory science with CSE architecture
    - Key concepts:
        - Forgetting curve and remembrance curve
        - Spacing engine and HLE implementation
        - Relay Cone and macro-forgetting
        - Signal Equation and mathematical unification
        - Scholar Continuum positioning
        - Six-axis integration framework
        - CSE implications
    - Outcome: Complete theoretical and practical mastery

MATERIAL DESCRIPTIONS

TEACHING DECKS

La Menara - The 12 Remarkable Relays (Part 1)

- **Type:** Teaching Deck (PowerPoint/PDF)
- **Size:** 9.2 MB
- **Duration:** 45–60 minutes
- **Audience:** All levels

- **Content:** Introduction to the 12 Relays, their historical significance, and civilisational progression
- **Key Sections:**
  - Relay 1: Fire (Energy & Control)
  - Relay 2: Tree (Agriculture & Sustainability)
  - Relay 3: River (Water & Transportation)
  - Relay 4: Stone (Structure & Permanence)
  - Relay 5: Road (Connection & Trade)
  - Relay 6: Metal (Materials & Tools)
  - Relay 7: Wheel (Motion & Mechanics)
  - Relay 8: Engine (Power & Acceleration)
  - Relay 9: Orbit (Scale & Space)
  - Relay 10: Web (Networks & Connection)
  - Relay 11: Code (Information & Intelligence)
  - Relay 12: Torus (Integration & Recursion)

Part 2: Civilisational Systems Engineering — 37 Base Case Examples

- **Type:** Teaching Deck (PowerPoint/PDF)
- **Size:** 27 MB
- **Duration:** 90–120 minutes
- **Audience:** Intermediate to advanced
- **Content:** Real-world infrastructure applications, case studies, and implementation patterns
- **Key Sections:**
  - 3–4 case studies per relay
  - Historical and contemporary examples
  - Failure modes and resilience strategies
  - Cross-relay dependencies
  - Implementation frameworks

RESEARCH PAPERS

CSE Foundational Paper

- **Type:** Academic Paper (PDF)
- **Size:** 459 KB
- **Duration:** 30–45 minutes
- **Audience:** All levels
- **Content:** Core theoretical framework and methodology for Civilisational Systems Engineering
- **Key Sections:**
  - Definition of CSE
  - System boundaries and relay mapping
  - Knowledge transmission mechanisms
  - Measurement frameworks
  - Integration with iAAi canon

UVM-DF Narrative Manuscript

- **Type:** Narrative Manuscript (PDF)
- **Size:** 548 KB
- **Duration:** 45–60 minutes
- **Audience:** Intermediate to advanced
- **Content:** Unified Value Model and Dearden Field narrative and implementation guide
- **Key Sections:**
  - Value integration framework
  - Field theory application
  - Narrative methodology
  - Implementation case studies
  - Dearden Field positioning in CSE

Hyper Learning Edutainment

- **Type:** Research Paper (PDF)
- **Size:** 210 KB
- **Duration:** 60–90 minutes
- **Audience:** Advanced practitioners, educators

- **Content:** Educational methodology combining CSE framework with immersive learning experiences
- **Key Sections:**
  - Spacing effect and memory science
  - HLE architecture and mechanics
  - Pedagogical implementation
  - Assessment frameworks
  - Scaling to civilisational levels

TP-064: What Ebbinghaus Could Not Build, iAAi Can (COMPLETE)

- **Type:** Turing Paper (PDF)
- **Size:** 169 KB
- **Duration:** 120–150 minutes
- **Audience:** Advanced scholars, system architects
- **Content:** Complete integration of memory science with CSE and iAAi architecture
- **Key Sections:**
  - Section 1: Ebbinghaus' forgetting curve
  - Section 2: Macro-forgetting and the Relay Cone
  - Section 3: Signal Equation unification
  - Section 4: 4Cs and 4Rs framework
  - Section 5: Remembrance curve
  - Section 6: Spacing Engine and HLE implementation
  - Section 7: Relay Cone compression and necessity
  - Section 8: Ebbinghaus in Scholar Continuum
  - Section 9: Six-axis integration
  - Section 10: CSE implications
  - Appendices: Definitions, visual plates, architecture, canon integration

VIDEO MATERIALS

CSE Framework Overview - Video 1

- **Type:** Video (MP4)

- **Size:** 2.4 MB
- **Duration:** 10 seconds
- **Audience:** All levels
- **Content:** Quick visual introduction to CSE principles and relay structure
- **Best Used:** Opening material or quick reference

CSE Application - Video 2

- **Type:** Video (MP4)
- **Size:** 2.5 MB
- **Duration:** 10 seconds
- **Audience:** All levels
- **Content:** Demonstration of CSE framework applied to real-world infrastructure
- **Best Used:** After Part 2 deck or as reinforcement

RECOMMENDED LEARNING PATHS

Path A: Quick Overview (1–2 hours)

1. CSE Framework Overview - Video 1 (10s)
2. La Menara - The 12 Remarkable Relays (45–60 min)
3. CSE Foundational Paper (30–45 min)
4. CSE Application - Video 2 (10s)

Outcome: Understanding of CSE basics and relay structure

Path B: Comprehensive Foundation (4–5 hours)

1. CSE Framework Overview - Video 1 (10s)
2. La Menara - The 12 Remarkable Relays (45–60 min)
3. CSE Foundational Paper (30–45 min)
4. Part 2: Civilisational Systems Engineering (90–120 min)
5. CSE Application - Video 2 (10s)

6. UVM-DF Narrative Manuscript (45–60 min)

Outcome: Practical understanding of CSE methodology and applications

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Path C: Complete Mastery (6–8 hours)

1. CSE Framework Overview - Video 1 (10s)
2. La Menara - The 12 Remarkable Relays (45–60 min)
3. CSE Foundational Paper (30–45 min)
4. Part 2: Civilisational Systems Engineering (90–120 min)
5. CSE Application - Video 2 (10s)
6. UVM-DF Narrative Manuscript (45–60 min)
7. Hyper Learning Edutainment (60–90 min)
8. TP-064: What Ebbinghaus Could Not Build, iAAi Can (120–150 min)

Outcome: Complete theoretical and practical mastery of CSE, memory science, and iAAi integration

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MATERIAL CORRELATIONS

Cross-References

Material	Correlates With	Connection
Part 1 Deck	TP-064 Sections 1–2	Relay structure and macro-forgetting
Part 2 Deck	TP-064 Section 10	CSE implications and applications
CSE Foundational Paper	TP-064 Sections 3–4	System definition and 4Cs/4Rs
UVM-DF Manuscript	TP-064 Section 9	Value integration and six-axis framework
Hyper Learning Edutainment	TP-064 Section 6	Spacing engine and HLE implementation
Video 1	All materials	Quick conceptual overview
Video 2	Part 2 Deck, TP-064 Section 10	Applied implementation

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USAGE GUIDELINES

For Educators

- **Tier 1 materials** are suitable for introductory courses (high school, first-year university)
- **Tier 2 materials** are suitable for intermediate courses (upper-level undergraduate, professional development)
- **Tier 3 materials** are suitable for advanced courses (graduate level, research seminars)
- **Videos** can be used as opening or closing material for any level

For Self-Directed Learners

- Follow one of the recommended learning paths (A, B, or C)

- Allow 15–20 minutes between materials for reflection
- Use videos as quick reference points
- Return to materials multiple times for deeper understanding

For Researchers

- Start with TP-064 for theoretical framework
- Use Part 2 Deck for case study examples
- Reference UVM-DF and HLE papers for specific methodologies
- Use CSE Foundational Paper for system definitions

MATERIAL STATUS & VERSIONING

Material	Version	Date	Status
La Menara - Part 1	1.0	2026	Complete
Part 2: CSE	1.0	2026	Complete
CSE Foundational Paper	1.0	2026	Complete
UVM-DF Narrative Manuscript	1.0	2026	Complete
Hyper Learning Edutainment	1.0	2026	Complete
TP-064 Complete	1.0	2026	Complete (Revised)
Video 1: Framework Overview	1.0	2026	Complete
Video 2: Application Demo	1.0	2026	Complete

FUTURE ADDITIONS

The following materials are planned for future inclusion:

- **TP-065:** Scholar Continuum update with Ebbinghaus integration
- **CSE Implementation Guide:** Step-by-step methodology for practitioners
- **Case Study Collection:** 50+ detailed infrastructure examples

- **Interactive Relay Explorer:** Web-based tool for relay navigation
- **Lecture Scripts:** Ready-to-use presentations for educators
- **Assessment Frameworks:** Evaluation tools for student mastery

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CONTACT & SUPPORT

For questions, feedback, or material requests:

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COLOPHON

Document: CSE Teaching Materials — Table of Contents **Version:** 1.0 **Date:** June 2026 **Status:** Complete **Canonical Alignment:** Full integration with iAAi canon, Relay Cone, Signal Equation, 4Cs/4Rs, Scholar Continuum

Key Thesis: Civilisational Systems Engineering is the applied discipline of engineering remembrance systems at civilisational scale, grounded in memory science (Ebbinghaus) and implemented through iAAi architecture.

Per Arya Ad Astra.

END OF CSE TEACHING MATERIALS TOC